Accessibility is a mere property: any two accessibility witnesses for the same element are equal

The Univalent Foundations Program

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Editorial note. The counted statement is the unlabelled author lem environment at source lines 928-930 and its proof environment opens on line 931, the next line; the pairing is by adjacency and the retained proof range 931-957 is the whole environment.

Editorial note. The environment carries no author label and the pinned book label-number dump has no entry for it, so no counter value is imposed and the counted lemma prints with the wrapper numbering; the original number is not claimed.

Editorial note. The counted statement and proof refer to no label outside the card: the retained ranges carry no cross reference command at all.

Editorial note. The author describes the type of the second coherence as monstrous but irrelevant; that sentence is retained verbatim, and the commented-out display of that type (source lines 946-952) is retained as a source comment, which renders nothing.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: accessibility is a mere property (unlabelled author lemma opening at line 928).

Allowed standard prerequisites (editorial).

- The accessibility family. Accessible elements of a set under a binary relation form an inductive family: an element is accessible as soon as every element below it is accessible, and any accessibility witness is built from a function choosing witnesses below.. The proof is an induction on one witness and then a second induction on the other, with the constructors of this family in both roles.
- Function extensionality. Two functions are equal as soon as they agree on every argument.. It reduces the final equality of two accessibility witnesses to a pointwise equality of their chosen witnesses below.

Lemma 0.0.1. *Accessibility is a mere property.*

Proof. We must show that for any $a : A$ and $s_1, s_2 : \mathsf{acc}(a)$ we have $s_1 = s_2$. We prove this by induction on s_1, with

$$P_1(a, s_1) :\equiv \prod_{s_2 : \mathsf{acc}(a)} (s_1 = s_2).$$

Thus, we must show that for any $a : A$ and $h_1 : \prod_{(b:A)}(b < a) \to \mathsf{acc}(b)$ and

$$k_1 : \prod_{(b:A)} \prod_{(l:b<a)} \prod_{(t:\mathsf{acc}(b))} h_1(b, l) = t,$$

we have $\mathsf{acc}_<(a, h) = s_2$ for any $s_2 : \mathsf{acc}(a)$. We regard this statement as $\prod_{(a:A)} \prod_{(s_2:\mathsf{acc}(a))} P_2(a, s_2)$, where

$$P_2(a, s_2) :\equiv \prod_{(h_1:\cdots)} \prod_{(k_1:\cdots)} (\mathsf{acc}_<(a, h_1) = s_2);$$

thus we may prove it by induction on s_2. Therefore, we assume $h_2 : \prod_{(b:A)}(b < a) \to \mathsf{acc}(b)$, and k_2 with a monstrous but irrelevant type, and must show that for any h_1 and k_1 with types as above, we have $\mathsf{acc}_<(a, h_1) = \mathsf{acc}_<(a, h_2)$. By function extensionality, it suffices to show $h_1(b, l) = h_2(b, l)$ for all $b : A$ and $l : b < a$. This follows from k_1. □